

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	✓ Deepa Dharshini	Reg. No.:	192525 213
Section / Batch:		Date:	21-07-26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q1. Consider the concept map: Algorithm → Growth Function → Big-Omega → Lower Bound → Best-case Explain how lower bound represents minimum execution time. Extend the map with performance guarantee node. Justify why best-case analysis is useful. Interpret how this contributes to overall performance understanding.

Q2. Consider the concept map: Recurrence Relation → Substitution Method → Expansion → Simplification → Final Complexity Trace the step-by-step transformation of recurrence into a closed form. Extend the map with Guess and Verification nodes. Justify how substitution helps solve non-standard recurrences. Interpret how this structured approach improves understanding.

Code of Conduct

I V. Deepa Dharshini

(192525213)

(Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Deepa**RUBRICS FOR CALCULATION**

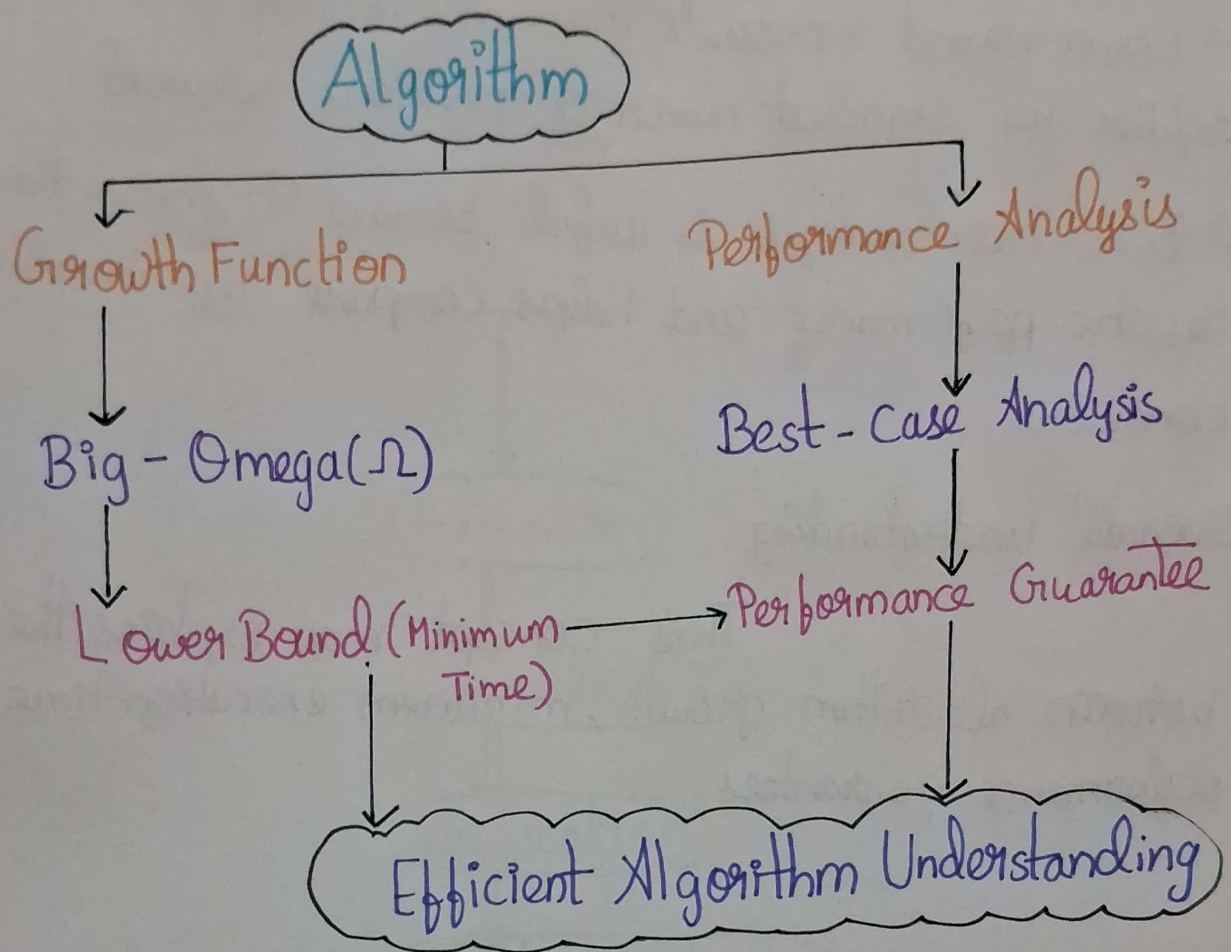
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Dr G Nagarajan		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Concept map:



Explanation:-

- Algorithm: A step by step method to solve a problem
- Growth Function: Shows how execution time changes as input size increases.
- Big-Omega(Ω): Represents the minimum time an algorithm needs.
- Lower bound: Indicates the least possible execution time, the algorithm can't perform faster than this
- Best-case Analysis: Measures performance under the most favorable input

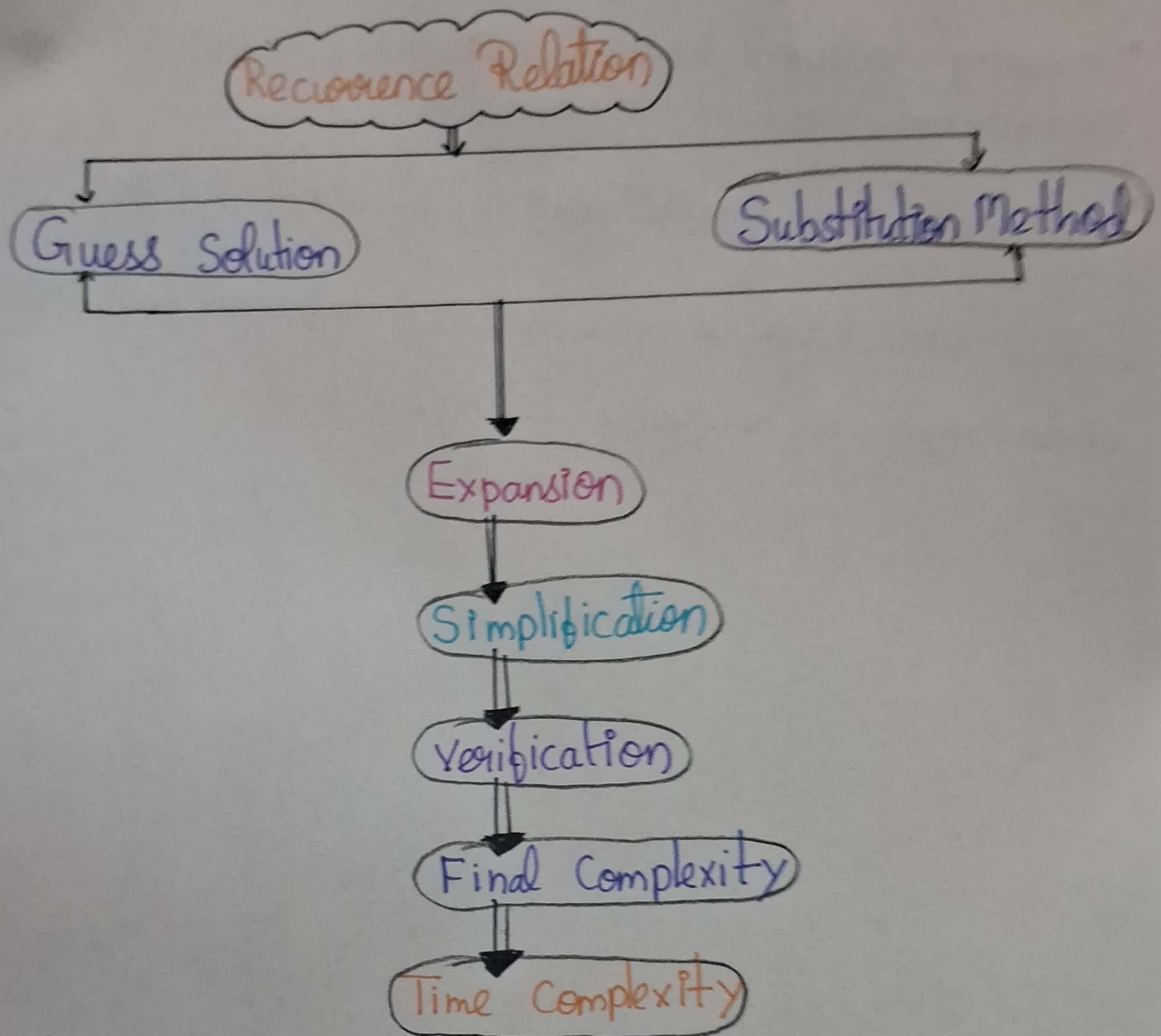
Justification

- Lower bound represents minimum execution time because it defines the smallest number of operations required.
- Best-case analysis is useful because it shows the fastest possible performance and helps compare algorithms under ideal conditions

Overall understanding:

This concept map explains the relationship between algorithm growth, minimum execution time, and performance guarantees.

Solving Recurrence Relation - Concept Map



Explanation:

- Recurrence Relation defines a problem in term of smaller instances
- Guess (Added) proposes a possible closed-form solution
- Substitution method replaces the recurrence with the guessed expression
- Expansion repeatedly unfolds the recurrence
- Simplification combines terms into a closed form
- Verification (Added) confirms the guessed solution satisfies relation
- Final complexity determines the algorithm's running time.

Justification

- Guessing followed by verification makes substitution effective even for non-standard recurrences
- Expansion and simplification reveal hidden patterns.
- The structured sequence reduces confusion and improves understanding of recurrence solving.